Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are:

(A) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{12}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(B) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

(C) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Q.2

If $0 < x < \pi$, and $\cos x + \sin x = 1/2$, then $\tan x$ is

(A) $(1 + \sqrt{7})/4$

(B) $(1 - \sqrt{7})/4$

(C) $(4 - \sqrt{7})/3$

(D) $-(4 + \sqrt{7})/3$

Q.3

The number of solutions of the equation $\sin 2x - 2\cos x + 4\sin x = 4$ in the interval $[0, 5\pi]$ is:

(A) 3

(B) 4

(C) 5

(D) 6

Q.4

If $0 \leq x < 2\pi$, then the number of real values of x which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is:

(A) 3

(B) 5

(C) 7

(D) 9

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Q.5

Number of values of $x \in [0, 2\pi]$ and satisfying the equation $\cos x \cos 2x \cos 3x = 1/4$ is:

(A) 6

(B) 8

(C) 20

(D) 24

Q.6

The solution set of the equation $\tan(\pi \tan x) = \cot(\pi \cot x)$ is

(A) $\{0\}$

(B) $\{\pi/4\}$

(C) ϕ

(D) none of these

Q.7

Sum of the root of the equation $2 \sin^2 \theta + \sin^2 2\theta = 2; 0 \leq \theta \leq \pi/2$ is

(A) $\pi/2$

(B) $3\pi/4$

(C) $7\pi/2$

(D) $5\pi/12$

Q.8

The number of solutions of the equation $\cos(\pi\sqrt{x-4}) \cos(\pi\sqrt{x}) = 1$ is

(A) 1

(B) 2

(C) more than two

(D) 0

Q.9

$\sin x + 2 \sin 2x = 3 + \sin 3x, 0 \leq x \leq 2\pi$ has

(A) 2 solutions in I quadrant

(B) one solution in II quadrant

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(C) no solution in any quadrant

(D) one solution in each quadrant

Q.10

The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$ where x is a variable, has real roots if p lies in the interval

(A) $(0, 2\pi)$

(B) $(-\pi, 0)$

(C) $(-\pi/2, \pi/2)$

(D) $(0, \pi)$

Q.11

The equation $\sin^4 x + \cos^4 x = a$ has a solution for

(A) all of values of a

(B) $a = -1$

(C) $a = -\frac{1}{2}$

(D) $\frac{1}{2} \leq a \leq 1$

Q.12

General solution of the equation
$(\sqrt{3} - 1)\sin\theta + (\sqrt{3} + 1)\cos\theta = 2$ is

(A) $2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

(B) $n\pi + (-1)^n \frac{\pi}{2}$

(C) $2n\pi \pm \frac{\pi}{4} - \frac{\pi}{12}$

(D) None

Q.13

If $\tan(\cot x) = \cot(\tan x)$, then

(A) $\sin 2x = \frac{2}{(2n+1)\pi}$

(B) $\sin x = \frac{4}{(2n+1)\pi}$

(C) $\sin 2x = \frac{4}{(2n+1)\pi}$

(D) None of these

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Q.14

If $\cos\theta + \cos 2\theta + \cos 3\theta = 0$, then the general value of θ is :

(A) $\theta = 2m\pi \pm 2\pi/3$

(B) $\theta = 2m\pi \pm \pi/4$

(C) $\theta = m\pi + (-1)^n 2\pi/3$

(D) $\theta = m\pi + (-1)^n \pi/3$

Q.15

The equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$
possesses a solution if

(A) $\mathrm{k} > -3$

(B) $\mathrm{k} < -2$

(C) $-3 \leq \mathrm{k} \leq -2$

(D) k is any positive integer

Answers & Solutions

Q.1 Answer:

$\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Solution:

We have

$$\sin\theta + \sin(4\theta) + \sin(7\theta) = 0$$
$$\Rightarrow 2\sin(4\theta)\cos(3\theta) + \sin(4\theta) = 0$$
$$\Rightarrow \sin(4\theta)[2\cos(3\theta) + 1] = 0$$
$$\Rightarrow \sin(4\theta) = 0 \text{ or } \cos(3\theta) = -1/2 = \cos(2\pi/3)$$
$$\Rightarrow \theta = \pi/4, \pi/2, 3\pi/4 \text{ or } 3\theta = 2\pi/3, 2\pi \pm 2\pi/3$$
$$\Rightarrow \theta = \frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{4\pi}{9}, \frac{3\pi}{4}, \frac{8\pi}{9}$$

Q.2 Answer:

$-(4+\sqrt{7})/3$

Solution:

$$2\sin x\cos x = (\sin x + \cos x)^2 - 1$$
$$= 1/4 - 1 = -3/4$$

As $0 < x < \pi, \sin x > 0$

Thus, $\cos x < 0$

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$\Rightarrow \quad \pi/2 < x < \pi$

Also, $(\sin x - \cos x)^2$

$= \sin^2 x + \cos^2 x - 2 \sin x \cos x$

$= 1 + 3/4 = 7/4$

$\Rightarrow \sin x - \cos x = \frac{\sqrt{7}}{2}$

Since $\sin x + \cos x = 1/2$

We get

$2 \sin x = (\sqrt{7} + 1)/2$

and $2 \cos x = (1 - \sqrt{7})/2$

$\therefore \quad \tan x = -\frac{\sqrt{7}+1}{\sqrt{7}-1} = -\frac{(\sqrt{7}+1)^2}{7-1}$

$= -\frac{1}{3}(4 + \sqrt{7})$

Q.3 Answer:

3

Solution:

$\sin 2x - 2 \cos x + 4 \sin x = 4$

$\Rightarrow \quad (2 \cos x + 4)(\sin x - 1) = 0$

$\Rightarrow \quad \sin x = 1 \quad \Rightarrow \quad x = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}$

Q.4 Answer:

7

Solution:

$\cos x + \cos 2x + \cos 3x + \cos 4x = 0$

$\Rightarrow (\cos 4x + \cos x) + (\cos 3x + \cos 2x) = 0$

$\Rightarrow 2 \cos \frac{5x}{2} \cos \frac{3x}{2} + 2 \cos \frac{5x}{2} \cos \frac{x}{2} = 0$

$\Rightarrow 2 \cos \frac{5x}{2} \left[\cos \frac{3x}{2} + \cos \frac{x}{2}\right] = 0$

$\Rightarrow 2 \cos \frac{5x}{2} \left[2 \cos x \cdot \cos \frac{x}{2}\right] = 0$

$\Rightarrow \cos \frac{x}{2} = 0, \cos x = 0$ **or** $\cos \frac{5x}{2} = 0$

$\Rightarrow \frac{x}{2} = \frac{\pi}{2}; x = \frac{\pi}{2}, \frac{3\pi}{2}; \frac{5x}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$

$\Rightarrow x = \pi, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{9\pi}{5}$

Q.5 Answer:

8

Solution:

$2(2 \cos 3x \cos x) \cos 2x = 1$

$\Rightarrow \quad 2(\cos 4x + \cos 2x) \cos 2x = 1$

$\Rightarrow \quad 2 \cos 4x \cos 2x = 1 - 2 \cos^2 2x = -\cos 4x$

$\Rightarrow \quad \cos 4x (2 \cos 2x + 1) = 0$

$\Rightarrow \quad \cos 4x = 0$ **or** $\cos 2x = -1/2$

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Now $\cos 4x = 0 \Rightarrow 4x = (2n+1)\pi/2, n \in \mathbf{I}$

$\Rightarrow x = (2n+1)\pi/8, n \in \mathbf{I}.$

and $0 \le (2n+1)\pi/4 \le 2\pi \Rightarrow 0 \le 2n+1 \le 8$

$\Rightarrow n = 0, 1, 2, 3$

Also, $\cos(2x) = -1/2 = \cos(2\pi/3)$

$\Rightarrow \quad 2x = 2n\pi \pm 2\pi/3, n \in \mathbf{I}$

$\Rightarrow \quad x = n\pi \pm \pi/3, n \in \mathbf{I}$

As $0 \le x \le 2\pi, x = \pi/3, 2\pi/3, 4\pi/3, 5\pi/3.$

Thus, there are 8 values of x**.**

Q.6 Answer:

ϕ

Solution:

$\tan(\pi \tan x) = \tan(\pi/2 - \pi \cot x)$

$\Rightarrow \quad \pi \tan x = \pi/2 - \pi \cot x \Rightarrow \tan x + \cot x = 1/2$

$\Rightarrow \quad 2\tan^2 x - \tan x + 2 = 0$

$\Rightarrow \quad \tan x = \frac{1 \pm \sqrt{1-16}}{4}$

which does not give real values of $\tan x$**.**

Q.7 Answer:

$3\pi/4$

Solution:

$4\sin^2\theta\cos^2\theta = 2\left(1 - \sin^2\theta\right)$

$\Rightarrow \quad \left(2\sin^2\theta - 1\right)\cos^2\theta = 0$

$\Rightarrow \quad \sin^2\theta = 1/2 \quad$ **or** $\quad \cos^2\theta = 0$

$\Rightarrow \theta = \pi/4 \quad$ **or** $\quad \theta = \pi/2$

Q.8 Answer:

1

Solution:

Given equation is possible if $\cos(\pi\sqrt{x-4}) = 1$ **and** $\cos(\pi\sqrt{x}) = 1.$

Since $\quad x - 4 \ge 0 \Rightarrow x \ge 4$ **and** $x \ge 0$

So $\quad x = 4$ **is the only solution.**

Q.9 Answer:

no solution in any quadrant

Solution:

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From the given relation we have

$\sin x - \sin 3x + 2\sin 2x = 3$

$\Rightarrow 2\sin x \cos 2x - 2\sin 2x + 3 = 0$

$\Rightarrow \quad (\sin x + \cos 2x)^2 + (\sin 2x - 1)^2 + 3$

$= \sin^2 x + \cos^2 2x + \sin^2 2x + 1$

$\Rightarrow \quad (\sin x + \cos 2x)^2 + (\sin 2x - 1)^2 + \cos^2 x = 0$

which is possible only if

$\sin x + \cos 2x = 0, \sin 2x = 1$ **and** $\cos x = 0$

which is not possible for any value of x**.**

Q.10 Answer:

$(0, \pi)$

Solution:

The equation has real roots if $\cos^2 p - 4(\cos p - 1)\sin p \geq 0$

or $\cos^2 p - 4\cos p \sin p + 4\sin p \geq 0$

or $(\cos p - 2\sin p)^2 - 4\sin^2 p + 4\sin p \geq 0$

or $(\cos p - 2\sin p)^2 + 4\sin p(1 - \sin p) \geq 0$

since $(\cos p - 2\sin p)^2 \geq 0, 1 - \sin p \geq 0$ **for all values of** p**, and for** $p \in (0, \pi), \sin p \geq 0$ **so that the discriminant is non-negative.**

Q.11 Answer:

$\frac{1}{2} \leq a \leq 1$

Solution:

The given equation can be written as $1 - 2\sin^2 x \cos^2 x = a$

$\Rightarrow \sin^2 2x = 2(1 - a) \Rightarrow 2(1 - a) \leq 1$

and $2(1 - a) \geq 0 \Rightarrow 1/2 \leq a \leq 1$

Q.12 Answer:

$2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

Solution:

Let $\sqrt{3} + 1 = r\cos\alpha$**, and** $\sqrt{3} - 1 = r\sin\alpha$

$\therefore r^2 = (\sqrt{3} + 1)^2 + (\sqrt{3} - 1)^2 = 8$ **i.e.** $\alpha = \pi/12$

From the equation, $r\cos(\theta - \alpha) = 2$

$\Rightarrow \quad \cos(\theta - \pi/12) = 1/\sqrt{2} = \cos(\pi/4)$

$\theta = 2n\pi \pm \pi/4 + \pi/12$

Q.13 Answer:

$\sin 2x = \frac{4}{(2n+1)\pi}$

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Solution:

$\tan(\cot x) = \cot(\tan x) = \tan\left(\frac{\pi}{2} - \tan x\right)$
$\Rightarrow \quad \cot x = n\pi + \frac{\pi}{2} - \tan x$
$[\because \tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha]$
$\Rightarrow \cot x + \tan x = n\pi + \frac{\pi}{2}$
$\Rightarrow \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} = (2n+1)\frac{\pi}{2}$
$\Rightarrow \frac{1}{\sin x \cos x} = (2n+1)\frac{\pi}{2}$
$\Rightarrow \frac{1}{\sin 2x} = \frac{(2n+1)\pi}{4}$
$\Rightarrow \quad \sin 2x = \frac{4}{(2n+1)\pi}$

Q.14 Answer:

$\theta = 2m\pi \pm 2\pi/3$

Solution:

Given $\cos\theta + \cos 2\theta + \cos 3\theta = 0$
$\Rightarrow (\cos 3\theta + \cos\theta) + \cos 2\theta = 0$
$\Rightarrow 2\cos 2\theta \cdot \cos\theta + \cos 2\theta = 0$
$\Rightarrow \cos 2\theta \cdot (2\cos\theta + 1) = 0$
we have, $\cos\theta = \cos\alpha \Rightarrow \theta = 2\mathrm{n}\pi \pm \alpha$
$\therefore$ **For general value of** θ, $\cos 2\theta = 0$
$\Rightarrow \cos 2\theta = \cos\frac{\pi}{2} \Rightarrow 2\theta = 2m\pi \pm \frac{\pi}{2}$
$\Rightarrow \theta = m\pi \pm \frac{\pi}{4}$ **or** $2\cos\theta + 1 = 0$
$\Rightarrow \cos\theta = \frac{-1}{2} \Rightarrow \cos\theta = \cos\frac{2\pi}{3}$
So, $\theta = 2m\pi \pm \frac{2\pi}{3}$

Q.15 Answer:

$-3 \leq \mathrm{k} \leq -2$

Solution:

We have,
$\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$
$\Rightarrow \quad \sin^2 x = \frac{(k+2) \pm \sqrt{(k+2)^2 + 4(k+3)}}{2}$
$= \frac{(k+2) \pm (k+4)}{2}$
$\Rightarrow \quad \sin^2 x = k+3 \ (\because \sin^2 x = -1$ **is not possible** $)$
Since $0 \leq \sin^2 x \leq 1, \therefore 0 \leq \mathrm{k}+3 \leq 1$
or $-3 \leq \mathrm{k} \leq -2$

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